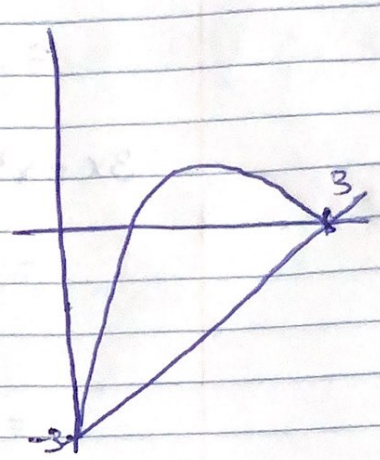
Quiz 1

① $y = x - 3$

$y = 4x - 3 - x^2$



$a = -3, \quad b = 3$

$$A = \int_{-3}^{3} (4x - 3 - x^2) - (x - 3)\, dx$$

$$A = \int_{-3}^{3} (4x - 3 - x^2 - x + 3)\, dx$$

$$A = \int_{-3}^{3} (3x - x^2)\, dx$$

$$A = \frac{3x^2}{2} - \frac{x^3}{3}\Big|_{-3}^{3}$$

$$A = \frac{3(3)^2}{2} - \frac{(3)^3}{3} - \frac{3(-3)^2}{2} + \frac{(-3)^3}{3}$$

$$A = \frac{27}{2} - \frac{27}{3} - \frac{27}{2} - \frac{27}{3}$$

$$A = -\frac{27}{3} - \frac{27}{3} = -\frac{54}{3} = -18$$